

Reading the Annual Layers: Dust Flux and Abrupt Transitions in a West Antarctic Ice Core

Klara Bjornstad, Emeka Nwosu, and Ruth Halloran
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Abstract

Atmospheric dust deposited in polar ice records the aridity and wind strength of distant continents. We present a high-resolution dust-flux profile from a newly recovered West Antarctic core spanning the last glacial termination. The record shows that the decline in southern-hemisphere dust at the end of the ice age was not gradual but stepped, with two abrupt drops separated by a millennial pause, implying that the atmospheric reorganisation of deglaciation proceeded in discrete jumps rather than a smooth ramp.

Introduction

Ice cores are layered diaries, and dust is one of their most legible inks. The quantity of wind-blown mineral dust trapped in each annual layer reflects how dry, windy, and exposed the source continents were when that snow fell. Reading a dust profile is therefore a way of reconstructing the circulation of an ancient atmosphere.

We report on a core recovered from a high-accumulation site in West Antarctica, chosen because rapid snowfall preserves fine temporal detail. Our focus is the last glacial termination, the transition out of the most recent ice age, when southern-hemisphere dust loading fell by roughly an order of magnitude.

Core Recovery and Dating

The core was drilled to bedrock and recovered in metre-long segments kept continuously frozen during transport. Annual layers were counted visually and confirmed by the seasonal cycling of chemical tracers, giving a timescale with uncertainty of a few decades across the interval of interest.

Dust was measured both as particle count, using a laser instrument scanning the melted core, and as the concentration of a characteristic crustal element. The two measures agreed closely, which gives us confidence that the structure we report is a feature of the atmosphere and not an artefact of one method.

A Stepped Decline

The central result is that the termination dust decline came in two distinct steps. An initial sharp drop reduced dust flux by about half over a span shorter than our dating resolution can fully separate, implying a change measured in decades to a century. This was followed by a plateau lasting close to a thousand years, during which dust held roughly steady.

A second, equally abrupt drop then brought dust flux down to its full interglacial baseline. The intervening plateau is the novel feature; earlier, lower-resolution records smeared the two steps into a single ramp and missed the pause entirely.

Implications for Deglacial Circulation

A stepped dust decline argues that the atmospheric reorganisation of deglaciation was threshold-like rather than linear. Each step plausibly marks the crossing of a tipping point in the position or strength of the southern westerly winds, which control how much dust reaches Antarctica from the arid continents upwind.

The millennial plateau between the steps suggests the system could rest in an intermediate state, neither fully glacial nor fully interglacial, for a long time before the second threshold was crossed. This has consequences for how we model the pace of past and future climate transitions, which too often assume smooth responses.

Conclusion

High-resolution dust data from West Antarctica reveal that the last deglaciation's atmospheric dust decline occurred in two abrupt steps bracketing a long pause. The finding underlines that smooth-looking climate transitions in coarse records may hide discrete jumps, and that resolution is not a luxury but a prerequisite for seeing how the climate system actually moves.

References

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